QUANT SERIES · PART 3 OF 4

Stochastic Calculus from Zero

Part 3: SDEs and pricing,
from Vasicek to the forward measure



Rates with Bip

STOCHASTIC CALCULUS FROM ZERO · PART 3 OF 4

Stochastic Calculus from Zero

Part 3: SDEs and pricing, with Python and QuantLib

Rates with Bip

Stochastic Calculus from Zero. Part 3: SDEs and pricing, with Python and QuantLib.

© 2026 Rates with Bip. This book is licensed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International licence (CC BY-NC-ND 4.0): you may copy and share it unchanged, for non-commercial purposes, crediting Rates with Bip. Licence: creativecommons.org/licenses/by-nc-nd/4.0. The companion notebooks are free under the MIT licence at github.com/RatesWithBip/curriculum.

Every path in this book is simulated with numpy and a fixed seed. The share (100, volatility 20%, risk-free rate 3.60%, real-world drift 8%, a call with strike 105 and a barrier at 130) and the short rate (3.85% today, long-run 3.50%, mean reversion 0.30, volatility 1%) are illustrative, made up for teaching. The one real series, the RBA cash rate target in Chapter 2, is from the Reserve Bank of Australia's Statistical Table F1 (Source: RBA 2026); the RBA does not endorse this book.

The history in this book is checked against the original publications through their Crossref records (Uhlenbeck and Ornstein 1930; Vasicek 1977; Cox, Ingersoll and Ross 1985; Feller 1951; Milstein 1975; Girsanov 1960; Kac 1949; Jamshidian 1989; Geman, El Karoui and Rochet 1995; Broadie, Glasserman and Kou 1997); sources are listed at the end of each chapter. Options are risky: a seller can lose far more than the premium received.

Every number in the text, tables and figures is computed by the companion Python notebooks (numpy, with QuantLib 1.43 as an independent check wherever a closed form exists), one per chapter, which reproduce each result. Every chapter's exercises are in its notebook too, and the code solutions check themselves.

This book is for education only. It is not financial, investment, legal or tax advice.

Contents

How to use this book 5

1. Stochastic Differential Equations 7

2. Mean Reversion 10

3. Square-Root Diffusions 14

4. Simulating SDEs 17

5. Girsanov's Theorem 20

6. Martingale Representation 24

7. Feynman-Kac 27

8. The Reflection Principle 30

9. Change of Numéraire 33

Exercises with answers 37

Checkpoint, glossary and further reading 48

How to use this book

This book follows the nine video episodes of Part 3 of *Stochastic Calculus from Zero* (Episodes 17 to 25), one chapter per episode, and goes further in each: the statements in full, and the short proofs the videos only sketch. It assumes Parts 1 and 2: conditional expectation, martingales and changes of odds on a coin tree, then Brownian motion, the Itô integral and the Itô-Doeblin formula. Part 3 uses those tools to write models down and to price with them.

One thread runs through the part: *make it a martingale, then take an expectation*. Chapters 1 to 4 write models as stochastic differential equations and simulate them. Chapter 5 chooses the measure that makes discounted prices martingales; Chapter 6 shows that such a price can be locked in by hedging; Chapter 7 turns the expectation into a PDE; Chapter 8 computes one in closed form by reflecting paths; Chapter 9 changes the unit of value to make a different ratio a martingale.

- **Chapter 1** says what an SDE is and solves geometric Brownian motion.
- **Chapters 2 and 3** solve and study two short-rate models, Vasicek and CIR, and compare them with the RBA's cash rate target.
- **Chapter 4** simulates SDEs with the Euler and Milstein schemes and measures two kinds of error.
- **Chapter 5** proves Girsanov's theorem for a constant drift change and prices a call with reweighted real-world paths.
- **Chapter 6** states the martingale representation theorem and tests it by hedging.
- **Chapter 7** proves the Feynman-Kac theorem and prices a Vasicek bond four ways.
- **Chapter 8** proves the reflection principle and prices a barrier option.
- **Chapter 9** changes numéraire, and explains why Black's formula works for bond options and caps.

Each chapter ends with its one idea to remember (*Bip's rule*) and a question answered at the start of the next chapter. The back matter has all forty-five exercises with full answers, the Part 3 checkpoint, a glossary and further reading.

Three checks for every price

Wherever a closed form exists, each result is computed at least two independent ways: the formula, QuantLib 1.43, a Monte Carlo simulation with its standard error, and sometimes a PDE or a different measure. When they agree, both the theory and the code have been tested.

Two running examples

The share is Series 3's: priced at 100, volatility 20%, a risk-free rate of 3.60% (continuous), a call with strike 105 expiring in a year, and an illustrative real-world drift of 8%. **The short rate** starts at 3.85%, is pulled towards 3.50% with mean reversion 0.30, and has volatility 1% a year. Both are made up for teaching. Every simulation uses the seed 2026, and Chapter 1 reuses Part 1's first Brownian path.

The notebooks are free at github.com/RatesWithBip/curriculum, in the `stochastic-calculus-from-zero` folder, each with an *Open in Colab* button. Each ends with its chapter's exercises, whose code solutions check themselves.

Reading alongside

Steven Shreve's *Stochastic Calculus for Finance II*, Chapters 5, 6 and 9 (and 7 for barriers), covers the same ground with full proofs; Bernt Øksendal's *Stochastic Differential Equations*, Chapters 5 and 8, is a compact alternative; Paul Glasserman's *Monte Carlo Methods in Financial Engineering*, Chapter 6, is the reference for discretisation; Brigo and Mercurio's *Interest Rate Models* for short-rate models. This book is written fresh from its own notebooks and uses none of their examples or exercises.

Free, and staying free

The videos, the notebooks and this book are free. If they help you, you can support the channel with a coffee at ko-fi.com/rateswithbip.

This book is for education only and is not financial advice. Options are risky: a seller can lose far more than the premium received.

CHAPTER 1 · VIDEO EPISODE 17

Stochastic Differential Equations

An SDE is a recipe for each small step: a drift plus scaled noise. Solve the one behind every share model, geometric Brownian motion, by guessing and checking with Itô's formula.

Before we start

The answers to the Part 2 checkpoint are at the back of the Part 2 book and in the Episode 16 notebook. This chapter uses one result from it constantly: the Itô-Doeblin formula,

$$df(t, X) = f_t\,dt + f_x\,dX + \tfrac{1}{2} f_{xx}\,(dX)^2, \qquad (dX)^2 = \sigma^2\,dt.$$

What an SDE is

A **stochastic differential equation** prescribes each small step of a process X:

$$dX_t = \mu(t, X_t)\,dt + \sigma(t, X_t)\,dW_t.$$

The first term is the **drift**, the second the **diffusion** term. Both may depend on time and on the current value. The shorthand means the integral equation

$$X_t = X_0 + \int_0^t \mu(s, X_s)\,ds + \int_0^t \sigma(s, X_s)\,dW_s,$$

where the second integral is the Itô integral of Part 2. A **strong solution** is an adapted process, built from the given Brownian motion W, with continuous paths, that satisfies this equation for every t.

Guess, apply Itô, check

Our share follows **geometric Brownian motion** (GBM), $dS = \mu S\,dt + \sigma S\,dW$.

Proposition. $S_t = S_0 \exp\left((\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\right)$ solves it.

Proof. Write $S = f(Y)$ with $f(y) = S_0 e^y$ and $Y_t = (\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t$, so $dY = (\mu - \tfrac{1}{2}\sigma^2)\,dt + \sigma\,dW$ and $(dY)^2 = \sigma^2\,dt$. Since $f' = f'' = f$, Itô-Doeblin gives

$$dS = S\,dY + \tfrac{1}{2}S\,(dY)^2 = S\big(\mu - \tfrac{1}{2}\sigma^2\big)\,dt + \sigma S\,dW + \tfrac{1}{2}\sigma^2 S\,dt,$$

which is $\mu S\,dt + \sigma S\,dW$. □

The $-\frac{1}{2}\sigma^2$ in the exponent exists to cancel the Itô term. Part 2 (Chapter 7) derived it from the other side: the log price drifts at $\mu - \frac{1}{2}\sigma^2$.

One path, two ways

On Episode 1's Brownian path (1,000 steps, $W_1 = 0.666$), compare the formula with simply stepping the SDE forward, $S_{n+1} = S_n(1 + \mu\,\Delta t + \sigma\,\Delta W_n)$ (the Euler scheme of Chapter 4):

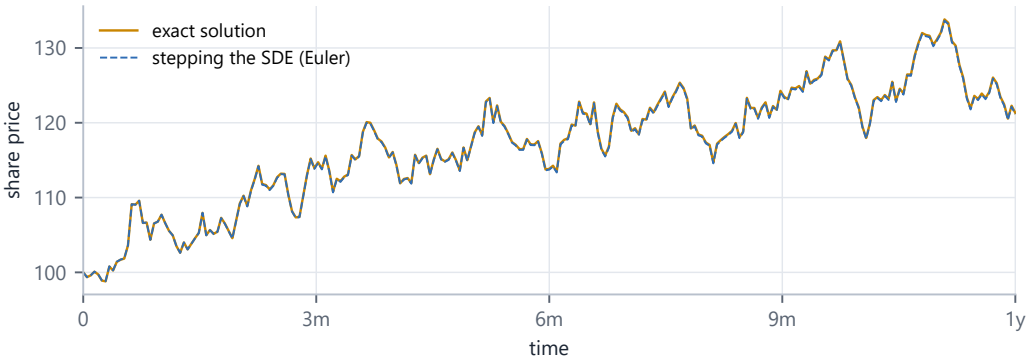


Figure 1.1. Formula and stepping almost coincide: the year ends at 121.32 and 121.17, and the largest gap along the way is 0.19.

Mean and median

Since $\ln S_T = \ln S_0 + (\mu - \frac{1}{2}\sigma^2)T + \sigma W_T$ is normal, S_T is **lognormal**. Using $E[e^{cW_T}] = e^{\frac{1}{2}c^2T}$,

$$E[S_T] = S_0 e^{(\mu - \frac{1}{2}\sigma^2)T}\, e^{\frac{1}{2}\sigma^2 T} = S_0 e^{\mu T}.$$

The median of S_T is e to the median of $\ln S_T$, which for a normal is its mean: $S_0 e^{(\mu - \frac{1}{2}\sigma^2)T}$. So the average grows at μ and the typical path at $\mu - \frac{1}{2}\sigma^2$.



Figure 1.2. 200,000 simulated year-end prices: mean 108.34 (theory 108.33), median 106.22 (theory 106.18). Only 46% of paths end above the mean.

When a solution exists

Theorem (existence and uniqueness). Suppose that for some constant K and all $t \le T$, x, y:

- $|\mu(t,x) - \mu(t,y)| + |\sigma(t,x) - \sigma(t,y)| \le K|x-y|$ (**Lipschitz**), and
- $|\mu(t,x)| + |\sigma(t,x)| \le K(1+|x|)$ (**linear growth**).

Then for each starting value there is exactly one strong solution on $[0,T]$, and $E[\sup_{t\le T} X_t^2] < \infty$.

The proof (Picard iteration, as for ordinary differential equations, with the Itô isometry in place of the usual integral bound) is in Øksendal, Chapter 5. GBM's μx and σx satisfy both conditions. The square root in Chapter 3's CIR model does not: $\sqrt{x}$ has infinite slope at 0, so CIR needs a separate argument (Yamada and Watanabe's).

A common mistake

"The drift in the SDE is the growth rate of the log price." It is the growth rate of the *average* price. The log price, and the median path, grow at $\mu - \frac{1}{2}\sigma^2$.

Bip's rule. Guess, apply Itô, check.

Try this. With 40% volatility and an 8% drift, what is the median price after a year?

Companion notebook: `s5e17_sdes_gbm.ipynb`. Every number in this chapter is computed by it.

CHAPTER 2 · VIDEO EPISODE 18

Mean Reversion

Interest rates don't wander off; they get pulled back. The Ornstein-Uhlenbeck SDE, Vasicek's short-rate model, builds the pull into the drift and has an exact, normal solution.

Last chapter's question

The answer is 100: the median is $100\,e^{0.08-\frac{1}{2}\times 0.16} = 100e^{0}$. Half the paths end below where they started, although the average grows by 8% a year.

The SDE

$$dr_t = a(\theta - r_t)\,dt + \sigma\,dW_t, \qquad a > 0.$$

Above the long-run level θ the drift is negative, below it positive, and the pull is proportional to the distance. This is the **Ornstein-Uhlenbeck** process (Uhlenbeck and Ornstein, 1930, for the velocity of a particle in a fluid). Vasicek (1977) used it for the short interest rate. Our illustrative rate starts at $r_0 = 3.85\%$, with long-run level $\theta = 3.50\%$, mean reversion $a = 0.30$ and volatility $\sigma = 1\%$ a year.

The exact solution

Proposition.

$$r_t = \theta + (r_0 - \theta)e^{-at} + \sigma \int_0^t e^{-a(t-s)}\,dW_s,$$

and r_t is normal with

$$E[r_t] = \theta + (r_0 - \theta)e^{-at}, \qquad \operatorname{Var}(r_t) = \frac{\sigma^2}{2a}\left(1 - e^{-2at}\right).$$

Proof. Apply Itô's product rule to $e^{at}r_t$. The factor e^{at} has no dW part, so there is no cross term:

$$d(e^{at}r_t) = ae^{at}r_t\,dt + e^{at}\,dr_t = a\theta e^{at}\,dt + \sigma e^{at}\,dW_t.$$

Integrate from 0 to t and multiply by e^{-at} to get the formula. The stochastic integral of the deterministic function $e^{-a(t-s)}$ is a limit of sums of independent normals, hence normal, with mean 0 (Part 2, Chapter 6) and, by the Itô isometry, variance $\sigma^2 \int_0^t e^{-2a(t-s)}\,ds = \sigma^2(1 - e^{-2at})/(2a)$. □

Because the transition from r_s to r_t is normal with known mean and variance, the notebook simulates it **exactly**, in steps of any size (`ou_step`).

Five years out

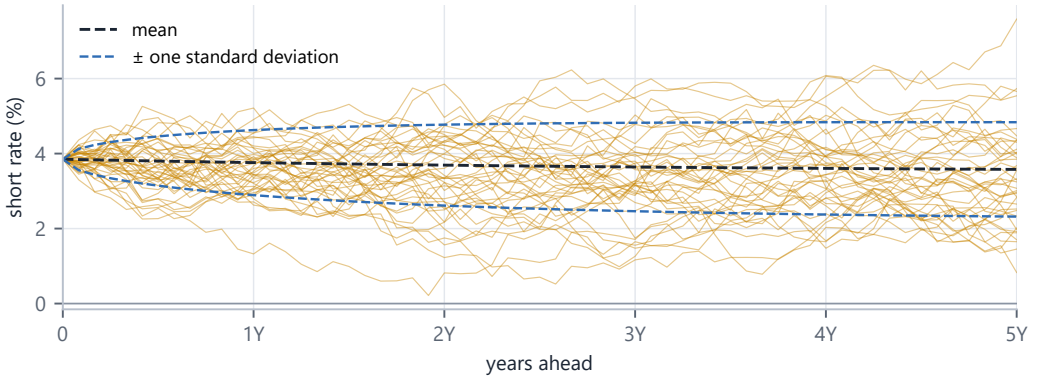


Figure 2.1. 40 of 100,000 paths with exact monthly steps, with the mean and one standard deviation either side. The mean slides towards θ; the spread grows, then stops growing.

METHOD	MEAN IN 5 YEARS	STANDARD DEVIATION
Formula	3.58%	1.26%
QuantLib `OrnsteinUhlenbeckProcess`	3.58%	1.26%
100,000 simulated paths	3.58%	1.26%

Half-life and long-run spread

The expected gap to θ decays like e^{-at}, so it halves every

$$\frac{\ln 2}{a} = 2.31 \text{ years}.$$

As $t \to \infty$ the variance tends to $\sigma^2/(2a)$: the long-run standard deviation is $\sigma/\sqrt{2a}$ = 1.29%. Without the pull, σW_t would already have a standard deviation of 2.24% after five years, and keep growing like $\sqrt{t}$. The limiting distribution, normal with mean θ and variance $\sigma^2/(2a)$, is **stationary**: a rate drawn from it stays distributed that way forever.

Negative rates

Because r_5 is normal, $P(r_5 < 0) = N(-E[r_5]/\text{sd}(r_5))$ = 0.22%; the simulation gives 0.22%. Whether that is a flaw depends on the market being modelled. Chapter 3's model rules it out.

Real policy rates move in steps

The Reserve Bank of Australia's cash rate target is set by its board and moves in steps:



Figure 2.2. The cash rate target, 4 Jan 2011 to 21 Sep 2026: 3,978 daily observations and 37 changes, between 0.10% and 4.75%. The longest run without a change lasted 717 trading days.

A model like Vasicek does not describe those decisions. It describes uncertainty about where the short rate might go, and is used to price bonds and options on them (Chapters 7 and 9). A realistic model of policy rates would add jumps on meeting dates: Part 4.

A common mistake

"Mean reversion makes rates predictable." The *mean* is predictable: 3.58% in five years. The rate is not: one standard deviation is still 1.26%.

Bip's rule. Pulled back, never pinned down.

Try this. If the pull doubles to $a = 0.6$, what is the half-life?

Sources

Companion notebook: `s5e18_mean_reversion.ipynb`. Every number in this chapter is computed by it.

1. American Physical Society (Crossref record), *G. E. Uhlenbeck and L. S. Ornstein, On the theory of the Brownian motion, Physical Review 36 (1930), 823-841*. https://doi.org/10.1103/PhysRev.36.823
2. QuantLib, *QuantLib 1.43: OrnsteinUhlenbeckProcess (expectation, stdDeviation)*. https://www.quantlib.org/
3. Reserve Bank of Australia, *Statistical Table F1 - Interest Rates and Yields, Money Market (daily): cash rate target*. https://www.rba.gov.au/statistics/tables/csv/f1-data.csv
4. Elsevier (Crossref record), *O. Vasicek, An equilibrium characterization of the term structure, Journal of Financial Economics 5 (1977), 177-188*. https://doi.org/10.1016/0304-405X(77)90016-2

CHAPTER 3 · VIDEO EPISODE 19

Square-Root Diffusions

Scale the noise by the square root of the rate and it fades near zero. Feller's condition says when zero is never reached; the mean is Vasicek's, the distribution is not.

Last chapter's question

The answer is 1.16 years: $\ln 2 / 0.6$. Double the pull, half the half-life.

The SDE

Cox, Ingersoll and Ross (1985) proposed

$$dr_t = a(\theta - r_t)\,dt + \sigma\sqrt{r_t}\,dW_t.$$

As $r \to 0$ the noise vanishes while the drift tends to $a\theta > 0$, pushing the rate back up. With the same a, θ and r_0 as Chapter 2, we choose $\sigma = 0.051$ so that the volatility at the start, $\sigma\sqrt{r_0}$, is the same 1%.

The coefficient $\sqrt{x}$ is not Lipschitz at 0, so Chapter 1's theorem does not apply directly; existence and uniqueness of a non-negative strong solution follow from Yamada and Watanabe's theorem, which only needs the volatility to be Hölder continuous of order $\frac{1}{2}$.

Feller's condition

Theorem (Feller, 1951). If $2a\theta \geq \sigma^2$, a CIR process started above zero never reaches zero. If $2a\theta < \sigma^2$, it reaches zero with positive probability, and is reflected back up.

Why the constant $2a\theta/\sigma^2$. Near zero, compare the push $a\theta\,dt$ with the noise, whose variance per unit time is $\sigma^2 r$. For a one-dimensional diffusion, zero is reached only if the **scale density** $s(x) = \exp\big(-\int^x 2\mu(y)/\sigma^2(y)\,dy\big)$ is integrable near 0. Here $2\mu/\sigma^2 = 2a(\theta - y)/(\sigma^2 y)$, so $s(x) \propto x^{-2a\theta/\sigma^2} e^{2ax/\sigma^2}$, and $\int_0 s(x)\,dx$ diverges exactly when $2a\theta/\sigma^2 \geq 1$.

Here $2a\theta = 0.021$ and $\sigma^2 = 0.0026$: met, easily. With $\sigma = 0.25$, $\sigma^2 = 0.0625$: broken. The boundary is $\sigma = \sqrt{2a\theta} = 0.145$.

Touching zero

A discrete step can overshoot below zero, where $\sqrt{r}$ is undefined. The **full-truncation** Euler scheme uses $r^+ = \max(r, 0)$ inside the drift and the square root:

$$r_{n+1} = r_n + a(\theta - r_n^+)\,\Delta t + \sigma\sqrt{r_n^+\,\Delta t}\;Z_n.$$

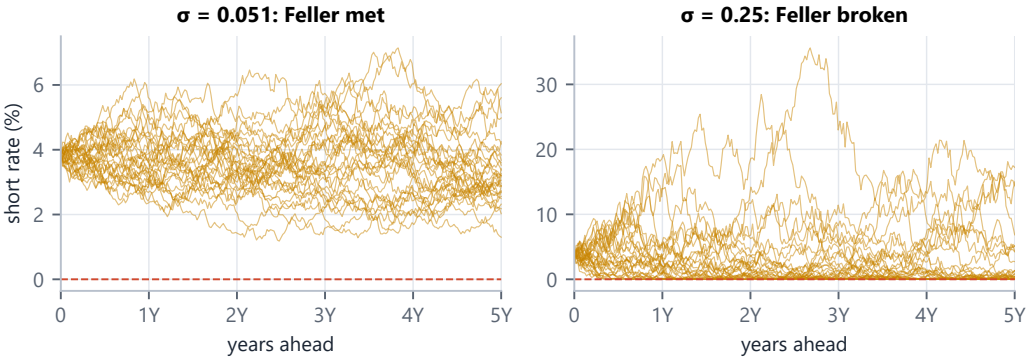


Figure 3.1. 30 of 20,000 paths each, five years of daily steps. With Feller's condition met, 0% touch zero; broken, 76.0% do, and bounce back.

Same mean, different shape

Proposition. The CIR mean is the same as Vasicek's: $E[r_t] = \theta + (r_0 - \theta)e^{-at}$.

Proof. Integrate the SDE and take expectations. The Itô integral $\int \sigma\sqrt{r}\,dW$ has mean zero, so $m(t) = E[r_t]$ satisfies $m(t) = r_0 + \int_0^t a(\theta - m(s))\,ds$, the same linear equation as in Chapter 2. □

The variance is also in closed form, $\mathrm{Var}(r_t) = r_0\frac{\sigma^2}{a}(e^{-at} - e^{-2at}) + \theta\frac{\sigma^2}{2a}(1 - e^{-at})^2$, and r_t follows a scaled non-central chi-squared distribution: skewed to the right, never negative.

FIVE YEARS OUT	MEAN	STANDARD DEVIATION
CIR, formula	3.58%	1.22%
CIR, simulated	3.58%	1.22%
Vasicek, formula	3.58%	1.26%

The lowest of the simulated CIR rates after five years is 0.45%.

Bond prices

A zero-coupon bond paying 1 in 5 years is worth $E[e^{-\int_0^5 r\,dt}]$ under the pricing measure (Chapters 5 and 7 explain why). CIR has an affine closed form, $A(\tau)e^{-B(\tau)r}$, like Vasicek's. It gives 0.832509, the same as QuantLib's `CoxIngersollRoss` model (0.832509); Vasicek gives 0.832532. The two models agree on the average path and differ in the tails.

Heston's variance

Series 3's Heston model lets a share's variance follow exactly this SDE, $dv = \kappa(\bar{v} - v)\,dt + \xi\sqrt{v}\,dW$. Feller's condition $2\kappa\bar{v} \geq \xi^2$ decides whether the variance can reach zero. Models calibrated to option prices often break it, which is why Heston simulations use schemes like full truncation.

A common mistake

"A model that keeps rates above zero is always better." If a market trades below zero, CIR cannot fit it at all. Choose the model for the market.

Bip's rule. Noise that fades near zero keeps rates above it.

Try this. Is Feller's condition met with $\sigma = 0.1$?

Sources

Companion notebook: `s5e19_cir.ipynb`. Every number in this chapter is computed by it.

1. The Econometric Society / JSTOR (Crossref record), *J. C. Cox, J. E. Ingersoll and S. A. Ross, A theory of the term structure of interest rates, Econometrica 53 (1985), 385-407*. https://doi.org/10.2307/1911242
2. Annals of Mathematics / JSTOR (Crossref record), *W. Feller, Two singular diffusion problems, Annals of Mathematics 54 (1951), 173-182*. https://doi.org/10.2307/1969318
3. QuantLib, *QuantLib 1.43: CoxIngersollRoss short-rate model (discountBond)*. https://www.quantlib.org/

CHAPTER 4 · VIDEO EPISODE 20

Simulating SDEs

Euler freezes the coefficients over each step; Milstein adds Part 2's Itô term. Milstein wins path by path, and buys nothing for prices.

Last chapter's question

The answer is yes: $\sigma^2 = 0.01$ is below $2a\theta = 0.021$.

Euler

Most SDEs have no formula, so we step them on a grid $t_n = n\,\Delta t$. Holding the coefficients at their values at the start of each step gives the **Euler** (Euler-Maruyama) scheme:

$$X_{n+1} = X_n + \mu(X_n)\,\Delta t + \sigma(X_n)\,\Delta W_n, \qquad \Delta W_n = W_{t_{n+1}} - W_{t_n}.$$

Over a step the true solution is $X_{t_n} + \int \mu(X_s)\,ds + \int \sigma(X_s)\,dW_s$. Replacing $\mu(X_s)$ by $\mu(X_n)$ costs little, but X_s moves by about $\sigma\,(W_s - W_{t_n})$, of order $\sqrt{\Delta t}$, so $\sigma(X_s)$ is already out of date.

Milstein

Derivation. Expand the volatility to first order along the step: $\sigma(X_s) \approx \sigma(X_n) + \sigma'(X_n)\,(X_s - X_n) \approx \sigma(X_n) + \sigma'\sigma\,(W_s - W_{t_n})$. Then

$$\int_{t_n}^{t_{n+1}} \sigma(X_s)\,dW_s \approx \sigma\,\Delta W_n + \sigma'\sigma \int_{t_n}^{t_{n+1}} (W_s - W_{t_n})\,dW_s,$$

and the last integral is Part 2's $\int W\,dW$ on one step: $\tfrac{1}{2}\big((\Delta W_n)^2 - \Delta t\big)$. So the **Milstein** scheme (Milstein, 1974) is

$$X_{n+1} = X_n + \mu\,\Delta t + \sigma\,\Delta W_n + \tfrac{1}{2}\,\sigma\sigma'\big((\Delta W_n)^2 - \Delta t\big).$$

For GBM, $\sigma(x) = \sigma x$, so $\sigma\sigma' = \sigma^2 x$.

Two kinds of error

On a grid of N steps up to T:

- **Strong error**: $E\,|X_N - X_T|$, how far each simulated path lands from the true path driven by the *same* Brownian motion. A scheme has **strong order** γ if this is at most $C\,\Delta t^{\gamma}$.
- **Weak error**: $|E[g(X_N)] - E[g(X_T)]|$ for smooth payoffs g, which is what a price needs. **Weak order** β means at most $C\,\Delta t^{\beta}$.

Under smoothness conditions, Euler has strong order $\frac{1}{2}$ and weak order 1; Milstein has strong order 1 and weak order 1.

Why Euler's strong order is only $\frac{1}{2}$. Euler drops $\frac{1}{2}\sigma\sigma'((\Delta W)^2 - \Delta t)$ at each step. These terms have mean zero, size Δt, and are independent across steps, so over $N = T/\Delta t$ steps they add up like $\sqrt{N}\,\Delta t = \sqrt{T\,\Delta t}$. *Why Milstein doesn't help prices:* the term it adds has mean zero, so it barely moves averages.

Measured

GBM with $\mu = 8\%$, $\sigma = 20\%$, 5,000 paths, each driven by one set of Brownian steps on a grid of 1,024, added up in blocks for the coarser grids. The exact solution uses the same draws, so the errors contain no sampling noise from comparing different paths.

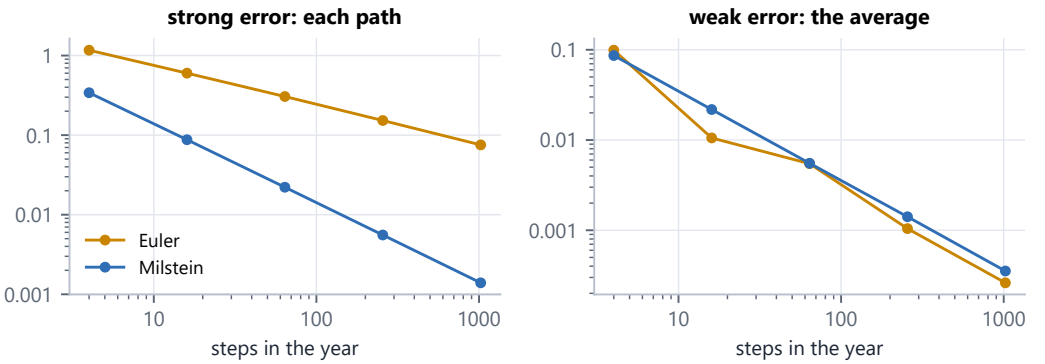


Figure 4.1. Left: strong error falls with slope -0.49 for Euler and -0.99 for Milstein on log-log axes. Right: the weak errors fall together, with slopes -1.02 and -0.99.

Strong: average |scheme − exact| at year end. Weak: |average of scheme − average of exact|, same draws.

STEPS	EULER, STRONG	MILSTEIN, STRONG	EULER, WEAK	MILSTEIN, WEAK
4	1.1679	0.3411	0.0990	0.0864
16	0.6009	0.0876	0.0105	0.0218
64	0.3058	0.0221	0.0055	0.0055
256	0.1527	0.0056	0.0010	0.0014
1,024	0.0754	0.0014	0.0003	0.0004

For Euler on GBM the weak error is known exactly: $E[S_N] = S_0(1+\mu\Delta t)^N$ against $S_0 e^{\mu T}$, a bias of 0.0855 with 4 steps. At 1,024 steps, Euler's strong error is 54 times Milstein's.

Which to use

- When the path matters (testing hedges, path-dependent payoffs, multilevel Monte Carlo): Milstein, or exact steps where a model has them (GBM, Vasicek).
- For prices of plain payoffs: the scheme's bias is usually smaller than the Monte Carlo noise; Euler with more paths is fine.
- With several Brownian motions, Milstein needs iterated integrals $\int\int dW^i\, dW^j$ that have no simple formula, which is a main reason Euler stays popular.

A common mistake

"A better scheme always gives a better price." Ask which error you care about first.

Bip's rule. Strong for paths, weak for prices.

Try this. Euler's strong error was 0.075 with 1,024 steps. How many steps to halve it?

Sources

Companion notebook: `s5e20_euler_milstein.ipynb`. Every number in this chapter is computed by it.

1. SIAM (Crossref record), *G. N. Mil'shtejn, Approximate integration of stochastic differential equations, Theory of Probability and Its Applications 19 (1975), 557-562 (translation of the 1974 Russian original)*. https://doi.org/10.1137/1119062

CHAPTER 5 · VIDEO EPISODE 21

Girsanov's Theorem

Reweight every Brownian path by an exponential of where it ends, and Brownian motion gains a drift while its volatility stays put. Choose the market price of risk, and the share grows at the risk-free rate.

Last chapter's question

The answer is 4,096: strong order $\frac{1}{2}$ means halving the error takes four times as many steps.

Changing the weights on paths

Part 1, Chapter 8 changed the odds on the coin world: each world's new weight was its old weight times a ratio Z, with $E_P[Z] = 1$, and averages under the new measure were $E_Q[X] = E_P[ZX]$. Here the worlds are Brownian paths, and the ratio is

$$Z_T = \frac{dQ}{dP} = \exp\big(-\theta W_T - \tfrac{1}{2}\theta^2 T\big).$$

It is positive, and $E_P[Z_T] = e^{\frac{1}{2}\theta^2 T} e^{-\frac{1}{2}\theta^2 T} = 1$ by the normal moment generating function. A path that ends high gets less weight when $\theta > 0$.

The theorem

Theorem (Girsanov, 1960). Under Q, the process $\widetilde{W}_t = W_t + \theta t$, $0 \le t \le T$, is a Brownian motion.

Proof for constant θ. $\widetilde{W}$ starts at 0 and has continuous paths; it remains to show its increments over $0 = t_0 < t_1 < \cdots < t_n = T$ are independent normals with variances Δt_i. Compute their joint moment generating function under Q. With $W_T = \sum \Delta W_i$ and $T = \sum \Delta t_i$,

$$E_Q\Big[e^{\sum u_i \Delta \widetilde{W}_i}\Big] = E_P\Big[e^{\sum (u_i - \theta)\Delta W_i}\Big]\, e^{\sum u_i \theta \Delta t_i - \frac{1}{2}\theta^2 T}.$$

The ΔW_i are independent $N(0, \Delta t_i)$ under P, so the first factor is $\prod e^{\frac{1}{2}(u_i-\theta)^2 \Delta t_i}$. Multiplying out,

$$\tfrac{1}{2}(u_i - \theta)^2 + u_i\theta - \tfrac{1}{2}\theta^2 = \tfrac{1}{2}u_i^2,$$

so the whole expression is $\prod e^{\frac{1}{2}u_i^2 \Delta t_i}$: the generating function of independent $N(0, \Delta t_i)$ variables. □

For a random, adapted θ_t the weight is $Z_T = \exp\big(-\int_0^T \theta\, dW - \frac{1}{2}\int_0^T \theta^2\, dt\big)$, and the theorem holds provided $E_P[Z_T] = 1$; **Novikov's condition**, $E_P\big[e^{\frac{1}{2}\int_0^T \theta_t^2\, dt}\big] < \infty$, is enough. Chapter 9 uses this version.

The market price of risk

Our share has $dS = \mu S\, dt + \sigma S\, dW$ under P. Choose

$$\theta = \frac{\mu - r}{\sigma} = \frac{8.0\% - 3.6\%}{20\%} = 0.22,$$

the **market price of risk**: the extra drift per unit of volatility. Substituting $dW = d\widetilde{W} - \theta\, dt$,

$$dS = \mu S\, dt + \sigma S\,(d\widetilde{W} - \theta\, dt) = rS\, dt + \sigma S\, d\widetilde{W}.$$

Under Q the share grows at the risk-free rate, so $e^{-rt}S_t$ is a Q-martingale: Q is the **risk-neutral measure**. The volatility is untouched.

Checked on simulated outcomes

400,000 real-world outcomes W_T, each with its weight Z_T:

QUANTITY	SIMULATED	THEORY
$E_P[Z_T]$	0.9998	1
$E_P[Z_T W_T] = E_Q[W_T]$	-0.219	-0.22
$\mathrm{Var}_Q(W_T + \theta T)$	0.999	1
$E_P[S_T]$	108.35	$100e^{0.08}$
$E_Q[S_T] = E_P[Z_T S_T]$	103.66	forward 103.67

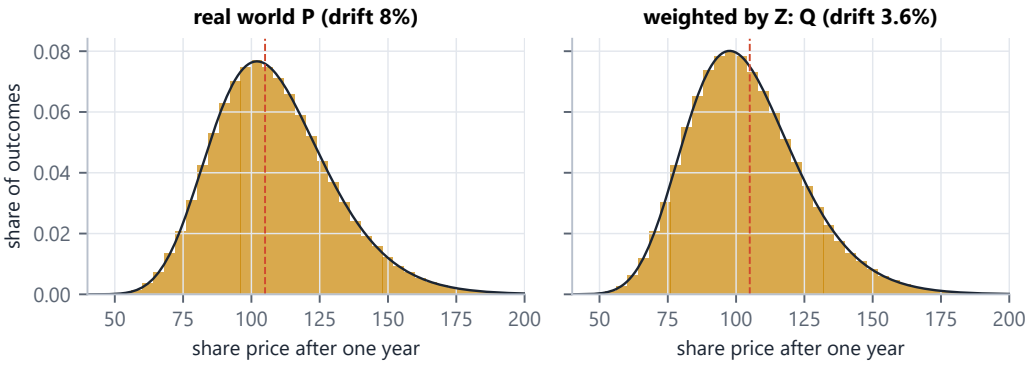


Figure 5.1. The same outcomes counted plainly (left) and with weights Z (right), against the lognormal densities with drift 8% and 3.6%. The dashed line is the strike, 105.

The price

The call pays $\max(S_T - K, 0)$ with $K = 105$:

METHOD	PRICE
Real-world average payoff, discounted	9.91
The same, weighted by Z_T	7.395 ± 0.016
Black-Scholes formula	7.390
QuantLib `AnalyticEuropeanEngine`	7.390

Discounting the real-world average gives the wrong price, because it prices in the share's real-world drift. Weighting by Z_T gives the Black-Scholes price within one standard error.

A common mistake

"The risk-neutral world has different paths." It has the same paths with new weights. And the volatility cannot change: it is read off each path from its quadratic variation, $\sum(\Delta \ln S)^2 \to \sigma^2 T$, an almost-sure limit, and P and Q agree on which events have probability one.

Bip's rule. Change the weights and the drift changes; the volatility doesn't.

Try this. If the drift equals the risk-free rate, what are θ and Z_T?

Sources

Companion notebook: `s5e21_girsanov.ipynb` . Every number in this chapter is computed by it.

1. SIAM (Crossref record), *I. V. Girsanov, On transforming a certain class of stochastic processes by absolutely continuous substitution of measures, Theory of Probability and Its Applications 5 (1960), 285-301*. https://doi.org/10.1137/1105027
2. QuantLib, *QuantLib 1.43: VanillaOption with AnalyticEuropeanEngine (BlackScholesMertonProcess)*. https://www.quantlib.org/

CHAPTER 6 · VIDEO EPISODE 22

Martingale Representation

Every martingale driven by a Brownian motion is an Itô integral against it. For an option's discounted price, the integrand is the hedge, and the price can be locked in.

Last chapter's question

The answer: $\theta = 0$, so $Z_T = e^0 = 1$. The real-world weights are already risk-neutral.

The theorem

Theorem (martingale representation). Let $\widetilde{W}$ be a Brownian motion under Q and $\mathcal{F}_t$ the information generated by it. If M is a martingale with respect to $\mathcal{F}_t$ with $E[M_T^2] < \infty$, there is an adapted H with $E\big[\int_0^T H_t^2\,dt\big] < \infty$ such that

$$M_t = M_0 + \int_0^t H_s\,d\widetilde{W}_s, \qquad 0 \le t \le T.$$

Part 2 (Chapter 6) showed that Itô integrals are martingales. This is the converse: in a world whose only randomness is one Brownian motion, there are no other martingales. The proof (via the density of exponential martingales $e^{\int h\,dW - \frac{1}{2}\int h^2 dt}$ among square-integrable payoffs) is in Øksendal, Chapter 4, and Shreve II, Section 5.3. It guarantees H exists but does not say how to find it; Itô's formula usually does.

The option's martingale

Let $V_t = E_Q\big[e^{-r(T-t)}\max(S_T - K, 0) \mid \mathcal{F}_t\big]$ be the call's price. By the tower property (Part 1), $e^{-rt}V_t$ is a Q-martingale, so $d(e^{-rt}V_t) = H_t\,d\widetilde{W}_t$. The discounted share is also one: $d(e^{-rt}S_t) = \sigma e^{-rt}S_t\,d\widetilde{W}_t$. Setting $\Delta_t = H_t/(\sigma e^{-rt}S_t)$,

$$d(e^{-rt}V_t) = \Delta_t\,d(e^{-rt}S_t).$$

Replication. Start with V_0, hold Δ_t shares, and keep the rest in the bank. The discounted value of this portfolio changes by $\Delta_t\,d(e^{-rt}S_t)$, exactly as the discounted option does, so it equals the option's value at every time, and at T pays $\max(S_T - K, 0)$. The price is not an opinion: anyone can lock it in.

For Black-Scholes, $V_t = v(t, S_t)$ with v the formula, and Itô's formula applied to $e^{-rt}v(t, S_t)$ shows the $d\widetilde{W}$ coefficient is $\sigma e^{-rt} S_t \, v_S$, so $\Delta_t = v_S = N(d_1)$: today 0.5144.

Delta, recovered from paths

The representation says that over a short step the discounted option moves by Δ times the discounted share, with nothing else of order $\sqrt{dt}$. Regressing one on the other over 200,000 simulated one-day moves from today gives a slope of 0.5145 against $N(d_1) = 0.5144$, and the share explains 99.9% of the variance of the option's moves.

Hedging in practice

Sell the call for 7.39, hold Δ shares, keep the rest in cash at 3.6%, and rebalance n times in the year. The share follows the *real-world* drift of 8%.

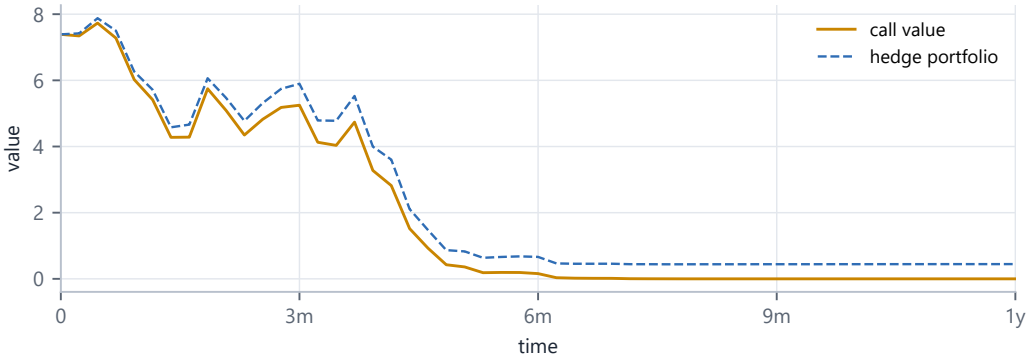


Figure 6.1. One path, rebalanced weekly: the hedge portfolio shadows the call all year and ends with 0.45 left over, from rebalancing only weekly.

Hedging error at expiry (portfolio minus payoff), 5,000 real-world paths each.

REBALANCING	TIMES A YEAR	ERROR: STANDARD DEVIATION	ERROR: AVERAGE
monthly	12	2.03	-0.027
weekly	52	1.01	-0.006
daily	252	0.45	-0.008

How fast the error shrinks

Over one rebalancing interval the hedged position gains $\frac{1}{2}\Gamma S^2\sigma^2\Delta t$ from the cash and option, and loses $\frac{1}{2}\Gamma S^2\sigma^2(\Delta W)^2$ (Itô's term) from the share's curvature: an error of $\frac{1}{2}\Gamma S^2\sigma^2\big(\Delta t - (\Delta W)^2\big)$, with mean zero and standard deviation of order Δt. Over $n =$

$T/\Delta t$ independent intervals these add up like $\sqrt{n}\,\Delta t \propto 1/\sqrt{n}$. From monthly to daily is 21 times more rebalancing; the error falls 4.54 times, against $\sqrt{21} = 4.58$.

Complete markets

With one Brownian motion and one traded risky asset, every payoff that depends on the path of W can be replicated, so it has exactly one price: the market is **complete**. Add a source of randomness that no traded asset carries, like jumps or a random volatility, and the representation needs a second integrand with nothing to trade against it: the market is **incomplete**, and prices are no longer unique (Part 4).

A common mistake

"You need the real-world drift to hedge." The hedging error has the same size whether the share drifts at 0% (daily error spread 0.45) or 20% (0.44). The drift cancels between $\Delta\, dS$ and dV.

Bip's rule. Every fair game is a trading strategy.

Try this. Rebalance four times as often: what happens to the hedging error?

Companion notebook: `s5e22_martingale_representation.ipynb`. Every number in this chapter is computed by it.

CHAPTER 7 · VIDEO EPISODE 23

Feynman-Kac

A discounted expected payoff solves a PDE, because a martingale has no drift. One Vasicek bond, priced four ways.

Last chapter's question

The hedging error halves: it shrinks like $1/\sqrt{n}$, so four times the rebalancing halves it.

The theorem

Theorem (Feynman-Kac). Let $dX = \mu(t,X)\,dt + \sigma(t,X)\,dW$, let $r(t,x)$ be a discount rate and g a payoff, and suppose

$$u(t,x) = E\Big[e^{-\int_t^T r(s,X_s)\,ds}\, g(X_T) \;\Big|\; X_t = x\Big]$$

is smooth enough ($C^{1,2}$). Then u solves

$$u_t + \mu\, u_x + \tfrac{1}{2}\sigma^2\, u_{xx} - r\, u = 0, \qquad u(T,x) = g(x).$$

Mark Kac proved it in 1949, motivated by Richard Feynman's path integrals (1948).

Proof. Let $D_t = e^{-\int_0^t r(s,X_s)\,ds}$, so $dD = -rD\,dt$. By the Markov property and the tower property (Part 1),

$$M_t = D_t\, u(t,X_t) = E\big[D_T\, g(X_T) \mid \mathcal{F}_t\big]$$

is a martingale. Itô-Doeblin and the product rule (no cross term, since D has no dW part) give

$$dM_t = D_t\big(u_t + \mu u_x + \tfrac{1}{2}\sigma^2 u_{xx} - ru\big)\,dt + D_t\,\sigma u_x\,dW_t.$$

A continuous martingale with a dt term would have a non-zero drift somewhere, so the bracket vanishes wherever X can go. □

The converse also holds: a smooth solution of the PDE with $u(T,\cdot) = g$, under growth conditions, *is* the expectation (apply Itô to $D_t u(t,X_t)$ and take expectations). So an expectation can be computed by solving a PDE, and a PDE by simulating paths. Series 3

used this once, to derive the Black-Scholes PDE.

The Vasicek bond

A zero-coupon bond paying 1 at T is $P(t,r) = E\big[e^{-\int_t^T r_s\,ds} \mid r_t = r\big]$, with Chapter 2's Vasicek rate as the state (under the pricing measure), the rate itself as the discount rate and $g = 1$:

$$P_t + a(\theta - r)P_r + \tfrac{1}{2}\sigma^2 P_{rr} - rP = 0, \qquad P(T,r) = 1.$$

Affine solution. Guess $P = A(\tau)e^{-B(\tau)r}$ with $\tau = T - t$. Then $P_r = -BP$, $P_{rr} = B^2P$ and $P_t = -(A'/A - B'r)P$. Dividing the PDE by P:

$$-\frac{A'}{A} + B'r - a\theta B + aBr + \tfrac{1}{2}\sigma^2 B^2 - r = 0.$$

The terms in r give $B' = 1 - aB$, $B(0) = 0$, so $B(\tau) = (1 - e^{-a\tau})/a$; the rest gives $(\ln A)' = -a\theta B + \tfrac{1}{2}\sigma^2 B^2$, $A(0) = 1$, which integrates to

$$\ln A(\tau) = \Big(\theta - \frac{\sigma^2}{2a^2}\Big)\big(B(\tau) - \tau\big) - \frac{\sigma^2 B(\tau)^2}{4a}.$$

A model whose bond prices are exponentials of affine functions of the state is called **affine**; CIR (Chapter 3) is another.

Four answers

For $T = 5$ years and $r_0 = 3.85\%$:

METHOD	PRICE OF 1 PAID IN 5 YEARS
Closed form (affine)	0.832532
QuantLib `Vasicek.discountBond`	0.832532
Monte Carlo, 50,000 paths	0.832567 ± 0.000147
PDE, Crank-Nicolson	0.832533

The Monte Carlo uses exact weekly steps and the trapezoid rule for $\int r$. The PDE is solved backwards from $P(T,r) = 1$ on a grid of rates from -10% to 20%; at the edges, far from today's rate, the P_r and P_{rr} terms are dropped.

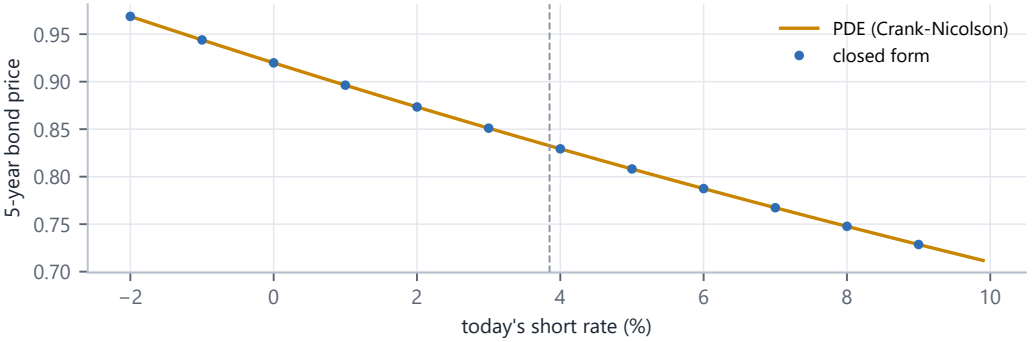


Figure 7.1. One PDE solve prices the bond for every starting rate on the grid; the dots are the closed form. Between −2% and 10% the largest gap is 8.4e-09.

Which road

With one or two state variables, a PDE grid is fast and gives every starting point at once. With many factors, or payoffs that depend on the whole path, grids grow exponentially with the dimension while Monte Carlo's error still falls like $1/\sqrt{N}$, so simulation wins.

A common mistake

"PDE pricing and Monte Carlo are different models." They are two algorithms for one number: one SDE, one measure, one payoff. If they disagree, one of them has a bug.

Bip's rule. Every expectation is a PDE, and every PDE an expectation.

Try this. With no volatility, the rate follows its mean exactly. What is the 5-year bond worth?

Sources

Companion notebook: `s5e23_feynman_kac.ipynb`. Every number in this chapter is computed by it.

1. American Mathematical Society (Crossref record), *M. Kac, On distributions of certain Wiener functionals, Transactions of the American Mathematical Society 65 (1949), 1-13*. https://doi.org/10.1090/S0002-9947-1949-0027960-X
2. QuantLib, *QuantLib 1.43: Vasicek short-rate model (discountBond)*. https://www.quantlib.org/

CHAPTER 8 · VIDEO EPISODE 24

The Reflection Principle

Mirror a Brownian path after it first hits a level, and the chance of ever reaching that level is twice the chance of ending above it. It prices barrier options, and shows why daily checks overprice knock-outs.

Last chapter's question

The answer is 0.8319. With $\sigma = 0$ the rate is $\theta + (r_0 - \theta)e^{-at}$, and the bond is $\exp\big(-\theta T - (r_0 - \theta)(1 - e^{-aT})/a\big)$. Volatility adds a little (Jensen: e^{-x} is convex).

Paths that care where they have been

Write $M_T = \max_{t \le T} W_t$ for the **running maximum**, and $\tau_m = \inf\{t : W_t = m\}$ for the first time W reaches a level $m > 0$; then $M_T \ge m$ exactly when $\tau_m \le T$.

The reflection principle

Theorem. For $m > 0$,

$$P(M_T \ge m) = 2\,P(W_T \ge m) = 2\big(1 - N(m/\sqrt{T})\big).$$

Proof. Split $\{M_T \ge m\}$ by where the path ends:

$$P(M_T \ge m) = P(M_T \ge m,\ W_T \ge m) + P(M_T \ge m,\ W_T < m).$$

The first event is just $\{W_T \ge m\}$, since a path ending above m has reached it. For the second, by the **strong Markov property** the path after τ_m, $B_s = W_{\tau_m + s} - m$, is a Brownian motion independent of what happened before. Replacing B by $-B$ (mirroring the path in the line $y = m$ after τ_m) gives a path with the same law, and it swaps ending below m with ending above it. So the two terms are equal. □

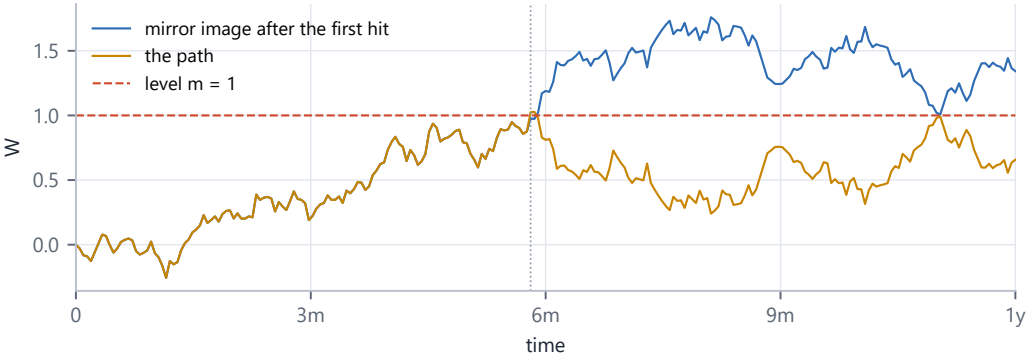


Figure 8.1. A simulated path first reaches 1 at t = 0.48 and ends at 0.66; its mirror image after that moment ends at 1.34. Both are equally likely.

For $m = 1, T = 1$: $2(1 - N(1)) = 31.73\%$, twice the 15.87% of paths that end above 1. In other words, M_T has the same distribution as $|W_T|$.

Grids miss crossings

METHOD	SHARE OF PATHS WITH MAXIMUM ≥ 1
Daily grid (252 steps)	30.2%
Fine grid (4,096 steps)	31.5%
Reflection principle	31.7%

20,000 paths for each grid. A path can cross the level and come back between two grid points, so a grid always undercounts.

A barrier option

An **up-and-out call** pays $\max(S_T - K, 0)$ only if the share never trades at or above the barrier B before expiry. Applying the reflection principle to $\ln S_t$, a Brownian motion with drift (handled by a Girsanov change of measure, Chapter 5), gives the joint law of $(S_T, \max S)$ and a closed form, implemented in QuantLib's `AnalyticBarrierEngine`. With $K = 105$ and $B = 130$, monitored continuously:

METHOD	PRICE
QuantLib, analytic	1.899
Monte Carlo, Brownian-bridge correction	1.888 ± 0.014
Monte Carlo, daily checks only	2.039
Vanilla call, no barrier	7.390

The Monte Carlo uses 100,000 risk-neutral paths with daily steps.

The Brownian-bridge correction

Proposition. Let X be a Brownian motion with volatility σ (any constant drift). Given $X_t = x$ and $X_{t+\Delta t} = y$, both below b, the probability that X crossed b in between is

$$p = \exp\Big(- \frac{2(b-x)(b-y)}{\sigma^2\,\Delta t} \Big).$$

Sketch. Conditional on its endpoints, the path is a **Brownian bridge**, and the drift drops out. Reflection in b maps bridges from x to y that touch b onto paths from x to $2b - y$, so p is the ratio of the normal densities of reaching $2b - y$ and y from x in time Δt:

$$p = \exp\Big(- \frac{(2b-y-x)^2 - (y-x)^2}{2\sigma^2 \Delta t} \Big) = \exp\Big(- \frac{2(b-x)(b-y)}{\sigma^2 \Delta t} \Big).$$

Weighting each simulated path by $\prod(1-p)$ over its steps gives an unbiased estimate of a continuously monitored knock-out from a daily grid. Checking only at the closes misses the crossings in between: here 20% of paths were knocked out at a daily close, too few, and the price came out 7.4% too high.

A common mistake

"Simulating daily is close enough for a continuously monitored barrier." It is not: use the bridge correction, or, for a contract that really is checked daily, the continuous formula with the barrier moved away by $e^{0.5826\,\sigma\sqrt{\Delta t}}$ (Broadie, Glasserman and Kou, 1997; Exercise 8.4).

Bip's rule. Reflect after the first hit.

Try this. What is the chance a Brownian motion reaches 2 within a year?

Sources

Companion notebook: `s5e24_reflection_barriers.ipynb`. Every number in this chapter is computed by it.

1. Wiley (Crossref record), *M. Broadie, P. Glasserman and S. Kou, A continuity correction for discrete barrier options, Mathematical Finance 7 (1997), 325-349*. https://doi.org/10.1111/1467-9965.00035
2. QuantLib, *QuantLib 1.43: BarrierOption with AnalyticBarrierEngine (continuously monitored barrier)*. https://www.quantlib.org/

CHAPTER 9 · VIDEO EPISODE 25

Change of Numéraire

Price in units of a zero-coupon bond, and under the matching forward measure the discount factor comes out of the average and forward prices become martingales. Why Black's formula works for bond options and caps.

Last chapter's question

The answer is 4.55%: $2(1 - N(2))$.

Numéraires

A **numéraire** is a traded asset with a positive price N_t, used as the unit of value.

Theorem (change of numéraire). Let Q be the risk-neutral measure, with bank account $B_t = e^{\int_0^t r\,ds}$, and N a numéraire. Define Q^N by

$$\frac{dQ^N}{dQ} = \frac{N_T / N_0}{B_T / B_0}.$$

Then for every traded price V, V_t / N_t is a Q^N-martingale, and in particular $V_0 = N_0\, E^N\left[V_T / N_T\right]$. (Geman, El Karoui and Rochet, 1995.)

Proof. N_t / B_t is a positive Q-martingale (a discounted traded price), so $Z_t = (N_t/N_0)/(B_t/B_0)$ is a valid density process with $E_Q[Z_T] = 1$. Note that $Z_t\, V_t / N_t = (B_0/N_0)\, V_t / B_t$. For $s < t$, the change-of-measure rule for conditional expectations (Part 1, Chapter 8) and the Q-martingale property of V/B give

$$E^N\left[\frac{V_t}{N_t}\,\middle|\,\mathcal{F}_s\right] = \frac{1}{Z_s}\,E_Q\left[Z_t \frac{V_t}{N_t}\,\middle|\,\mathcal{F}_s\right] = \frac{B_0}{N_0 Z_s}\,\frac{V_s}{B_s} = \frac{V_s}{N_s}. \qquad \square$$

The forward measure

With $N_t = P(t,T)$, the zero-coupon bond maturing at T, the measure is the **T-forward measure** Q^T. Since $P(T,T) = 1$,

$$V_0 = P(0,T)\, E^T[V_T].$$

Under Q the price is $E_Q[e^{-\int_0^T r} V_T]$, with a random discount factor *inside* the average. Under Q^T the discount factor comes *out*, as today's bond price.

The drift under Q^T. In Vasicek the bond's volatility is $-\sigma B(t,T)$, with $B(t,T) = (1 - e^{-a(T-t)})/a$: $dP/P = r\,dt - \sigma B(t,T)\,dW$. So $dZ/Z = -\sigma B(t,T)\,dW$, a Girsanov density with $\theta_t = \sigma B(t,T)$ (Chapter 5, random-θ version), and $W_t^T = W_t + \int_0^t \sigma B(s,T)\,ds$ is a Q^T-Brownian motion. Substituting into $dr = a(\theta - r)\,dt + \sigma\,dW$:

$$dr = \big(a(\theta - r) - \sigma^2 B(t,T)\big)\,dt + \sigma\,dW^T.$$

Paths with high rates, where the bond is cheap, count for less.

A bond option, three ways

A call expiring at $T_1 = 1$ on the zero-coupon bond maturing at $T_2 = 5$, struck at the **forward bond price**

$$K = F_0 = \frac{P(0,T_2)}{P(0,T_1)} = \frac{0.83253}{0.96270} = 0.86479.$$

At T_1 the bond is worth $P(T_1,T_2) = A(T_2 - T_1)e^{-B(T_2-T_1)r_{T_1}}$ (Chapter 7), so only r_{T_1} needs simulating.

METHOD	PRICE (BASIS POINTS OF FACE)
QuantLib `Vasicek.discountBondOption`	67.09
Risk-neutral paths, each discounted by $e^{-\int r}$	67.23 ± 0.32
T_1-forward-measure paths, discounted once by $P(0,T_1)$	67.14 ± 0.31

100,000 paths each, daily steps.

The forward is a martingale

$F_t = P(t,T_2)/P(t,T_1)$ is a traded price in units of the numéraire $P(t,T_1)$, so it is a Q^{T_1}-martingale, and at T_1 it equals $P(T_1,T_2)$. So $E^{T_1}[P(T_1,T_2)] = F_0$. On the forward-measure paths, $P(1,5)$ averages 0.86475 ± 0.00006 against $F_0 = 0.86479$. Under Q the matching statement needs the discount factor inside: on the risk-neutral paths, $E_Q[e^{-\int_0^1 r} P(1,5)]$ comes out at 0.83251, against $P(0,5) = 0.83253$.

Why Black's formula works

In Vasicek, $\ln F_{T_1}$ is normal under Q^{T_1} (it is affine in the normal r_{T_1}), with variance

$$v^2 = \sigma^2 B(T_1, T_2)^2 \, \frac{1 - e^{-2aT_1}}{2a}, \qquad v = 2.02\%.$$

A lognormal martingale with known total variance is exactly the setting of Black's formula, so

$$C = P(0, T_1)\big(F_0 N(d_1) - K N(d_2)\big), \qquad d_{1,2} = \frac{\ln(F_0/K) \pm \tfrac{1}{2}v^2}{v}.$$

This is Jamshidian's (1989) bond option formula; it gives 67.09 basis points, QuantLib's price exactly. Caps work the same way. Each forward rate is a martingale under the forward measure of its payment date. Assume it lognormal and you get Black-76; assume it normal and you get Bachelier's formula (Series 3).

A common mistake

"With random rates you can take today's discount factor outside the average." Only under the forward measure of the payment date, with the drift changed accordingly. Under the risk-neutral measure the discount factor and the payoff are correlated, and both must stay inside.

Part 3 in one page

CHAPTER	IDEA
1	SDEs; GBM solved by guess, Itô, check
2, 3	Short rates: Vasicek (normal), CIR (Feller)
4	Euler and Milstein: strong for paths, weak for prices
5	Girsanov: reweight paths, change the drift
6	Martingale representation: the price can be hedged
7	Feynman-Kac: expectations are PDEs
8	Reflection: barriers in closed form
9	Numéraires: the forward measure and Black's formula

Bip's rule. Choose the numéraire that makes your problem a martingale.

Try this. The Part 3 checkpoint is at the back of the book.

Sources

Companion notebook: `s5e25_forward_measure.ipynb`. Every number in this chapter is computed by it.

1. Cambridge University Press / JSTOR (Crossref record), *H. Geman, N. El Karoui and J.-C. Rochet, Changes of numéraire, changes of probability measure and option pricing, Journal of Applied Probability 32 (1995), 443-458*. https://doi.org/10.2307/3215299
2. Wiley (Crossref record), *F. Jamshidian, An exact bond option formula, Journal of Finance 44 (1989), 205-209*. https://doi.org/10.1111/j.1540-6261.1989.tb02413.x
3. QuantLib, *QuantLib 1.43: Vasicek short-rate model (discountBondOption)*. https://www.quantlib.org/

Exercises with answers

Five exercises per chapter: three by hand and two in code. They are the same exercises as at the end of each chapter's notebook, where the code solutions run and check themselves; try each before reading its answer. "Section" in an exercise means a section of that chapter's notebook.

Chapter 1: Stochastic Differential Equations

Exercise 1.1 *(by hand)*. Solve the arithmetic SDE $dX = \mu\,dt + \sigma\,dW$, $X_0 = x$. What are the mean and variance of X_t?

Answer. Integrate both sides: $X_t = x + \mu t + \sigma W_t$. Mean $x + \mu t$, variance $\sigma^2 t$. (This is Bachelier's model.)

Exercise 1.2 *(by hand)*. Show that $S_t = S_0 \exp(\mu t + \sigma W_t)$ (no $-\frac{1}{2}\sigma^2$) solves a different SDE. Which one?

Answer. Itô with $Y = \mu t + \sigma W$: $dS = S\,dY + \frac{1}{2}S\sigma^2\,dt = (\mu + \frac{1}{2}\sigma^2)S\,dt + \sigma S\,dW$. Its drift is $\mu + \frac{1}{2}\sigma^2$, not μ.

Exercise 1.3 *(by hand)*. Why is the median of S_T equal to $S_0 e^{(\mu - \frac{1}{2}\sigma^2)T}$?

Answer. S_T is an increasing function of $\ln S_T$, so its median is $e^{\text{median of } \ln S_T}$. $\ln S_T$ is normal, so its median is its mean, $\ln S_0 + (\mu - \frac{1}{2}\sigma^2)T$.

Exercise 1.4 *(in code)*. Step the SDE with 10, 100 and 1,000 steps along Episode 1's path (add up its steps in blocks). How does the gap to the exact solution change?

Solution 1.4, from the notebook

```
gaps = {}
for n in (10, 100, 1_000):
    d = dw.reshape(n, -1).sum(axis=1)
    s = S0
    for x in d:
        s *= 1 + MU / n + VOL * x
    gaps[n] = s - exact[-1]
assert abs(gaps[1_000]) < abs(gaps[10])
print({n: round(float(g), 3) for n, g in gaps.items()}, "(fewer steps, bigger gap)")
```

What it prints. `{10: 0.731, 100: -0.443, 1000: -0.152} (fewer steps, bigger gap)`

Exercise 1.5 *(in code)*. What is the real-world chance that the share ends the year above 100? Compute $N\big((\mu - \tfrac{1}{2}\sigma^2)/\sigma\big)$ and check it on the 200,000 simulated ends.

Solution 1.5, from the notebook

```
p = float(norm_cdf((MU - 0.5 * VOL ** 2) / VOL))
sim = float((ends > S0).mean())
assert abs(p - sim) < 0.005
print(f"formula {p:.4f}, simulated {sim:.4f}")
```

What it prints. `formula 0.6179, simulated 0.6183`

Chapter 2: Mean Reversion

Exercise 2.1 *(by hand)*. Use the integrating factor e^{at} to solve $dX = -aX\,dt + \sigma\,dW$ with $X_0 = x$.

Answer. $d(e^{at}X) = \sigma e^{at}\,dW$, so $X_t = xe^{-at} + \sigma\int_0^t e^{-a(t-s)}\,dW_s$: mean xe^{-at}, variance $\sigma^2(1 - e^{-2at})/(2a)$.

Exercise 2.2 *(by hand)*. What is the long-run (stationary) standard deviation of the Vasicek rate, and how does it compare with Brownian motion's after 5 years?

Answer. $\sigma/\sqrt{2a} = 0.01/\sqrt{0.6} = 1.29\%$, against $\sigma\sqrt{5} = 2.24\%$ for σW_5: mean reversion caps the spread.

Exercise 2.3 *(by hand)*. Starting at $r_0 = 3.85\%$, when has the expected rate covered 90% of the distance to $\theta = 3.5\%$?

Answer. $e^{-at} = 0.1$, so $t = \ln 10/0.3 = 7.7$ years.

Exercise 2.4 *(in code)*. Simulate with the Euler scheme $(r + a(\theta - r)\Delta t + \sigma\sqrt{\Delta t}\,Z)$, monthly, and compare the 5-year mean and standard deviation with the exact ones.

Solution 2.4, from the notebook

```
rng_e = np.random.default_rng(1)
x = np.full(100_000, R0)
for _ in range(60):
    x = x + A * (THETA - x) / 12 + SIG * math.sqrt(1 / 12) * rng_e.standard_normal(x.size)
assert abs(x.mean() - m5) < 2e-4 and abs(x.std() / sd5 - 1) < 0.02
print(f"Euler: mean {x.mean():.4%}, sd {x.std():.4%}; exact {m5:.4%}, {sd5:.4%}")
```

What it prints. `Euler: mean 3.5810%, sd 1.2648%; exact 3.5781%, 1.2584%`

Exercise 2.5 *(in code)*. Estimate a from one long simulated path: regress $r_{t+\Delta t}$ on r_t. The slope should be $e^{-a\Delta t}$.

Solution 2.5, from the notebook

```
rng_f = np.random.default_rng(2)
path = [R0]
for _ in range(12 * 200):                                  # 200 years of monthly rates
    path.append(float(ou_step(path[-1], 1 / 12, A, THETA, SIG, rng_f.standard_normal())))
path = np.array(path)
slope = np.polyfit(path[:-1], path[1:], 1)[0]
a_hat = -math.log(slope) * 12
assert abs(a_hat - A) < 0.1
print(f"estimated a = {a_hat:.3f} (true {A}); even 200 years of data pin it down only
roughly")
```

What it prints. `estimated a = 0.378 (true 0.3); even 200 years of data pin it down only roughly`

Chapter 3: Square-Root Diffusions

Exercise 3.1 *(by hand).* Is Feller's condition met for $a = 0.5$, $\theta = 4\%$, $\sigma = 0.15$? And for $\sigma = 0.25$?

Answer. $2a\theta = 0.04$. $\sigma^2 = 0.0225 \le 0.04$: met. $\sigma^2 = 0.0625 > 0.04$: not met, zero can be reached.

Exercise 3.2 *(by hand).* Show that the CIR mean $m(t) = E[r_t]$ satisfies $m' = a(\theta - m)$, the same equation as Vasicek's.

Answer. Integrate the SDE and take expectations: $E[\int \sigma\sqrt{r}\, dW] = 0$ (an Itô integral is a martingale), so $m(t) = r_0 + \int_0^t a(\theta - m(s))\, ds$. Differentiate.

Exercise 3.3 *(by hand).* Apply Itô to $X = \sqrt{r}$. What is the volatility of X?

Answer. $f' = \frac{1}{2\sqrt{r}}$, $f'' = -\frac{1}{4}r^{-3/2}$, $(dr)^2 = \sigma^2 r\, dt$: $dX = \left(\frac{a\theta - \sigma^2/4}{2X} - \frac{a}{2}X\right)dt + \frac{\sigma}{2}\, dW$. The square root has constant volatility $\sigma/2$; the drift explodes upwards near 0 when $a\theta > \sigma^2/4$.

Exercise 3.4 *(in code).* Find the fraction of paths that touch zero within 5 years for σ from 0.05 to 0.25. Where does it start rising?

Solution 3.4, from the notebook

```
rows = []
for s in (0.05, 0.10, 0.145, 0.20, 0.25):
    _, h, _ = cir_paths(s, n_paths=5_000)
    rows.append({"sigma": s, "sigma_sq": s * s, "2a theta": 2 * A * THETA, "touched zero":
float(h.mean())})
assert rows[0]["touched zero"] == 0 and rows[-1]["touched zero"] > 0.5
print(pd.DataFrame(rows).to_string(index=False))
print("Feller boundary: sigma =", round(math.sqrt(2 * A * THETA), 3), "(daily steps blur it a
little)")
```

What it prints. `sigma sigma_sq 2a theta touched zero; 0.050 0.002500 0.021 0.0000; 0.100 0.010000 0.021 0.0002; 0.145 0.021025 0.021 0.1198; 0.200 0.040000 0.021 0.5528; 0.250 0.062500 0.021 0.7580; Feller boundary: sigma = 0.145 (daily steps blur it a little)`

Exercise 3.5 *(in code).* Price the 5-year CIR bond by Monte Carlo: average $e^{-\int_0^5 r\,dt}$ over simulated paths, and compare with the closed form.

Solution 3.5, from the notebook

```
rng_b = np.random.default_rng(3)
n, dt = 20_000, 1 / 252
x = np.full(n, R0); integral = np.zeros(n)
for _ in range(5 * 252):
    xp = np.maximum(x, 0)
    xn = x + A * (THETA - xp) * dt + SIG * np.sqrt(xp * dt) * rng_b.standard_normal(n)
    integral += 0.5 * (np.maximum(x, 0) + np.maximum(xn, 0)) * dt
    x = xn
mc = np.exp(-integral)
se = mc.std() / math.sqrt(n)
assert abs(mc.mean() - bond["closed"]) < 3 * se + 1e-4
print(f"Monte Carlo {mc.mean():.5f} ± {se:.5f}; closed form {bond['closed']:.5f}")
```

What it prints. `Monte Carlo 0.83252 ± 0.00023; closed form 0.83251`

Chapter 4: Simulating SDEs

Exercise 4.1 *(by hand).* Write the Milstein scheme for GBM, $dS = \mu S\,dt + \sigma S\,dW$.

Answer. $\sigma(x) = \sigma x$, $\sigma'(x) = \sigma$: $S_{n+1} = S_n\big(1 + \mu\Delta t + \sigma\Delta W + \tfrac{1}{2}\sigma^2((\Delta W)^2 - \Delta t)\big)$.

Exercise 4.2 *(by hand).* For $dX = \mu\,dt + \sigma\,dW$ with constants, what does Milstein add to Euler?

Answer. Nothing: $\sigma' = 0$. Euler is then exact on the grid points, for both strong and weak error.

Exercise 4.3 *(by hand)*. Why is Euler's strong order only $\frac{1}{2}$? (Hint: the size of the term it drops.)

Answer. It drops $\frac{1}{2}\sigma\sigma'((\Delta W)^2 - \Delta t)$, of size Δt per step. These errors have mean zero and are independent, so over $N = T/\Delta t$ steps they add up like $\sqrt{N}\,\Delta t = \sqrt{T\Delta t}$.

Exercise 4.4 *(in code)*. Apply both schemes to the CIR model of Episode 19 ($\sigma(r) = \sigma\sqrt{r}$, so $\sigma\sigma' = \sigma^2/2$) with 12 steps a year for 5 years. How often does each go negative?

Solution 4.4, from the notebook

```
R0, THETA, A, SIG = 0.0385, 0.035, 0.30, 0.25        # Feller broken, to make negatives pos-
sible
rng_c = np.random.default_rng(4)
n, dt = 50_000, 1 / 12
e = np.full(n, R0); m = np.full(n, R0); neg_e = np.zeros(n, bool); neg_m = np.zeros(n, bool)
for _ in range(60):
    dW = math.sqrt(dt) * rng_c.standard_normal(n)
    ep, mp = np.maximum(e, 0), np.maximum(m, 0)
    e = e + A * (THETA - ep) * dt + SIG * np.sqrt(ep) * dW
    m = m + A * (THETA - mp) * dt + SIG * np.sqrt(mp) * dW + 0.25 * SIG ** 2 * (dW ** 2 - dt)
    neg_e |= e < 0; neg_m |= m < 0
assert neg_m.mean() < neg_e.mean()
print(f"went negative: Euler {neg_e.mean():.1%}, Milstein {neg_m.mean():.1%}")
```

What it prints. `went negative: Euler 75.7%, Milstein 75.5%`

Exercise 4.5 *(in code)*. Price the call (strike 105) under the risk-neutral drift 3.6% with Euler at 4, 16 and 64 steps, on the same draws as the exact solution. How fast does the price error fall?

Solution 4.5, from the notebook

```
R, K = 0.036, 105.0
ex = S0 * np.exp((R - 0.5 * VOL ** 2) + VOL * Z.sum(axis=1))
exact_price = math.exp(-R) * np.maximum(ex - K, 0).mean()
errs = []
for n in (4, 16, 64):
    dW = Z.reshape(P, n, NFINE // n).sum(axis=2)
    e = np.full(P, S0)
    for i in range(n):
        e = e + R * e / n + VOL * e * dW[:, i]
    errs.append(abs(math.exp(-R) * np.maximum(e - K, 0).mean() - exact_price))
assert errs[-1] < errs[0]
print("price error:", [round(float(x), 4) for x in errs], "(it shrinks as the steps grow)")
```

What it prints. `price error: [0.0655, 0.0062, 0.0025] (it shrinks as the steps grow)`

Chapter 5: Girsanov's Theorem

Exercise 5.1 *(by hand)*. Show $E_P[Z_T] = 1$, using $E[e^{cW_T}] = e^{\frac{1}{2}c^2T}$.

Answer. $E[e^{-\theta W_T}]e^{-\frac{1}{2}\theta^2 T} = e^{\frac{1}{2}\theta^2 T}e^{-\frac{1}{2}\theta^2 T} = 1$.

Exercise 5.2 *(by hand)*. Show $E_Q[W_T] = -\theta T$, so $E_Q[W_T + \theta T] = 0$. (Hint: $E_P[We^{cW}]$ is the derivative of the MGF in c.)

Answer. $E_P[W_T e^{cW_T}] = \frac{d}{dc}e^{\frac{1}{2}c^2T} = cTe^{\frac{1}{2}c^2T}$. With $c = -\theta$ and the factor $e^{-\frac{1}{2}\theta^2 T}$: $E_Q[W_T] = -\theta T$.

Exercise 5.3 *(by hand)*. Why can't a change of measure change the volatility?

Answer. Volatility is read off the quadratic variation, $\sum(\Delta S)^2 \to \int \sigma^2 S^2\, dt$, which is a pathwise limit. Q and P agree on which events have probability 1 (they are equivalent), so the limit is the same under both.

Exercise 5.4 *(in code)*. Using the weights, check that under Q the log return $\ln(S_T/S_0)$ has mean $(r - \frac{1}{2}\sigma^2)T$ and variance $\sigma^2 T$.

Solution 5.4, from the notebook

```
L = np.log(ST / S0)
qm = float((Z * L).mean()); qv = float((Z * L ** 2).mean() - qm ** 2)
assert abs(qm - (R - 0.5 * VOL ** 2) * T) < 0.002 and abs(qv / (VOL ** 2 * T) - 1) < 0.02
print(f"Q mean {qm:.4f} (theory {(R - 0.5 * VOL ** 2) * T:.4f}), Q variance {qv:.4f} (theory
{VOL ** 2 * T:.4f})")
```

What it prints. `Q mean 0.0162 (theory 0.0160), Q variance 0.0400 (theory 0.0400)`

Exercise 5.5 *(in code)*. Redo the price with a real-world drift of 15%. What is θ, and does the Z-weighted price change?

Solution 5.5, from the notebook

```
mu2 = 0.15
th2 = (mu2 - R) / VOL
W2 = np.random.default_rng(5).standard_normal(n)
S2 = S0 * np.exp((mu2 - 0.5 * VOL ** 2) * T + VOL * W2)
Z2 = np.exp(-th2 * W2 - 0.5 * th2 ** 2 * T)
p2 = Z2 * math.exp(-R * T) * np.maximum(S2 - K, 0)
se2 = p2.std() / math.sqrt(n)
assert abs(p2.mean() - price["bs"]) < 3 * se2
print(f"theta {th2:.2f}; weighted price {p2.mean():.3f} ± {se2:.3f}, Black-Scholes
{price['bs']:.3f}: the drift drops out")
```

What it prints. `theta 0.57; weighted price 7.404 ± 0.011, Black-Scholes 7.390: the drift drops out`

Chapter 6: Martingale Representation

Exercise 6.1 *(by hand)*. Write $e^{-rT}S_T$ as $M_0 + \int_0^T H_t\, d\widetilde{W}_t$. What is H, and what is the hedge?

Answer. $d(e^{-rt}S_t) = \sigma e^{-rt} S_t\, d\widetilde{W}_t$, so $M_0 = S_0$, $H_t = \sigma e^{-rt} S_t$. The hedge is $\Delta = 1$: hold one share.

Exercise 6.2 *(by hand)*. Why is the average hedging error close to zero even though the share drifts at 8%, not 3.6%?

Answer. The hedged portfolio's value changes by $\Delta\, dS - dV + r(\cdot)\, dt$; the dW and $\mu\, dt$ terms cancel between $\Delta\, dS$ and dV (Black-Scholes derivation). What is left over each step depends on $(\Delta W)^2 - \Delta t$, whose mean is 0 under any drift.

Exercise 6.3 *(by hand)*. A trader rebalances monthly and sees an error spread of 2.0. Roughly how often must they rebalance to get it to 0.5?

Answer. Error $\propto 1/\sqrt{n}$: a factor of 4 smaller needs 16 times as often, $12 \times 16 = 192$ times a year (about every business day and a bit).

Exercise 6.4 *(in code)*. Hedge (daily) with the wrong volatility, 15% instead of the true 20%. What happens to the mean error?

Solution 6.4, from the notebook

```
e_wrong = hedge(252, vol_hedge=0.15, rng=np.random.default_rng(9))
cheap = float(bs_call(S0, K, T, R, 0.15)[0])
assert e_wrong.mean() < -1
print(f"sold for {cheap:.2f} instead of {price0:.2f}; mean error {e_wrong.mean():.2f}, sd
{e_wrong.std():.2f}: "
      "the call was sold too cheap for the volatility the share delivered")
```

What it prints. `sold for 5.40 instead of 7.39; mean error -2.12, sd 1.01: the call was sold too cheap for the volatility the share delivered`

Exercise 6.5 *(in code)*. Hedge daily with real-world drift 0% and then 20%. Does the drift matter?

Solution 6.5, from the notebook

```
res = {m: hedge(252, mu=m, rng=np.random.default_rng(10)) for m in (0.0, 0.20)}
for m, e in res.items():
    print(f"drift {m:.0%}: mean error {e.mean():+.3f}, sd {e.std():.3f}")
assert abs(res[0.0].std() - res[0.20].std()) < 0.1
out["drift"] = {"zero": float(res[0.0].std()), "twenty": float(res[0.20].std())}
```

What it prints. `drift 0%: mean error -0.009, sd 0.451` ; `drift 20%: mean error -0.021, sd 0.441`

Chapter 7: Feynman-Kac

Exercise 7.1 *(by hand)*. Repeat the martingale argument without discounting: if $u(t,x) = E[g(X_T) \mid X_t = x]$, which PDE does u solve?

Answer. $u(t, X_t)$ is itself a martingale; its dt term must vanish: $u_t + \mu u_x + \frac{1}{2}\sigma^2 u_{xx} = 0$, $u(T,x) = g(x)$ (the backward Kolmogorov equation).

Exercise 7.2 *(by hand)*. Substitute $P = A(\tau)e^{-B(\tau)r}$ into the Vasicek PDE (with $P_t = -P_\tau$). Show that $B' = 1 - aB$, $B(0) = 0$, and solve it.

Answer. $P_r = -BP$, $P_{rr} = B^2 P$, $P_t = -P_\tau = -(A'/A - B'r)P$. Divide the PDE by P: $-A'/A + B'r - a\theta B + aBr + \frac{1}{2}\sigma^2 B^2 - r = 0$. The terms in r give $B' + aB - 1 = 0$, so $B' = 1 - aB$ and, with $B(0) = 0$, $B(\tau) = (1 - e^{-a\tau})/a$. The rest gives $(\ln A)' = -a\theta B + \frac{1}{2}\sigma^2 B^2$.

Exercise 7.3 *(by hand)*. Why does adding volatility ($\sigma > 0$) make the bond worth more than the deterministic one?

Answer. e^{-x} is convex, so by Jensen $E[e^{-\int r}] \geq e^{-E[\int r]}$, and $E[\int r]$ is the same with or without σ in Vasicek. Here the gain is small (see §8).

Exercise 7.4 *(in code)*. Refine the PDE grid (101, 201, 401 rate points) and watch the error to the closed form fall.

Solution 7.4, from the notebook

```
errs = []
for pts in (101, 201, 401):
    g_, p_ = pde_bond(TB, points=pts)
    errs.append(abs(float(np.interp(R0, g_, p_)) - closed))
assert errs[-1] < errs[0]
print([f"{e:.1e}" for e in errs])
```

What it prints. `['3.5e-06', '1.4e-06', '3.5e-07']`

Exercise 7.5 *(in code)*. Price a 10-year bond three ways: closed form, QuantLib, PDE.

Solution 7.5, from the notebook

```
c10 = float(vasicek_bond(10.0, R0, A, THETA, SIG))
q10 = ql.Vasicek(R0, A, THETA, SIG).discountBond(0.0, 10.0, R0)
g10, p10 = pde_bond(10.0, time_steps=1000)
d10 = float(np.interp(R0, g10, p10))
assert abs(c10 - q10) < 1e-10 and abs(d10 - c10) < 2e-4
print(f"10-year bond: closed {c10:.6f}, QuantLib {q10:.6f}, PDE {d10:.6f}")
```

What it prints. `10-year bond: closed 0.698985, QuantLib 0.698985, PDE 0.698986`

Chapter 8: The Reflection Principle

Exercise 8.1 *(by hand)*. What is the chance that a Brownian motion reaches 0.5 within a year?

Answer. $2(1 - N(0.5)) = 2 \times 0.3085 = 0.617$.

Exercise 8.2 *(by hand)*. What is the chance that W stays below 1 for four years?

Answer. $1 - 2(1 - N(1/\sqrt{4})) = 1 - 2(1 - N(0.5)) = 0.383$.

Exercise 8.3 *(by hand)*. Why is an up-and-in call plus an up-and-out call (same strike, same barrier) worth exactly the vanilla call?

Answer. On every path exactly one of them is alive at expiry and it pays $\max(S_T - K, 0)$; the other pays 0. Same payoff on every path, same price.

Exercise 8.4 *(in code)*. Broadie, Glasserman and Kou (1997): a barrier checked every Δt is priced well by the continuous formula with the barrier moved away by $e^{0.5826\,\sigma\sqrt{\Delta t}}$. Compare with the daily Monte Carlo.

Solution 8.4, from the notebook

```
shifted = BARRIER * math.exp(0.5826 * VOL * math.sqrt(dt))
bgk = ql_barrier(ql.Barrier.UpOut, shifted)
assert abs(bgk - mc["daily"]) < 3 * mc["daily_se"] + 0.01
print(f"shifted barrier {shifted:.2f}: formula {bgk:.4f}; daily Monte Carlo
{mc['daily']:.4f} ± {mc['daily_se']:.4f}")
```

What it prints. `shifted barrier 130.96: formula 2.0648; daily Monte Carlo 2.0387 ± 0.0146`

Exercise 8.5 *(in code)*. Check in-out parity with QuantLib: up-and-in plus up-and-out equals the vanilla call.

Solution 8.5, from the notebook

```
ui = ql_barrier(ql.Barrier.UpIn, BARRIER)
assert abs(ui + barrier["ql_up_out"] - barrier["vanilla"]) < 1e-8
print(f"up-and-in {ui:.4f} + up-and-out {barrier['ql_up_out']:.4f} = {ui +
barrier['ql_up_out']:.4f}; vanilla {barrier['vanilla']:.4f}")
```

What it prints. `up-and-in 5.4907 + up-and-out 1.8989 = 7.3896; vanilla 7.3896`

Chapter 9: Change of Numéraire

Exercise 9.1 *(by hand).* Under Q the bond has volatility $-\sigma B(t,T)$: $dP/P = r\,dt - \sigma B(t,T)\,dW$. Use Girsanov to show that under Q^T the Vasicek rate has drift $a(\theta - r) - \sigma^2 B(t,T)$.

Answer. $Z_t = P(t,T)/(B_t P(0,T))$ is a Q-martingale with $dZ/Z = -\sigma B(t,T)\,dW$, so it is a Girsanov weight with $\theta_t = \sigma B(t,T)$: $W^T = W + \int_0^t \sigma B(s,T)\,ds$ is a Q^T-Brownian motion. Substitute $dW = dW^T - \sigma B\,dt$ into $dr = a(\theta - r)\,dt + \sigma\,dW$.

Exercise 9.2 *(by hand).* Why is $P(t,T_2)/P(t,T_1)$ a martingale under Q^{T_1}, and what is its value at $t = T_1$?

Answer. $P(t,T_2)$ is a traded price and $P(t,T_1)$ is the numéraire; prices in units of the numéraire are martingales. At T_1, $P(T_1,T_1) = 1$, so it equals $P(T_1,T_2)$, and its expectation is today's forward $P(0,T_2)/P(0,T_1)$.

Exercise 9.3 *(by hand).* Show put-call parity for bond options: $C - P = P(0,T_2) - K\,P(0,T_1)$.

Answer. $C - P$ pays $P(T_1,T_2) - K$ at T_1. Under Q^{T_1}: value $= P(0,T_1)\big(E^{T_1}[P(T_1,T_2)] - K\big) = P(0,T_1)(F_0 - K) = P(0,T_2) - KP(0,T_1)$.

Exercise 9.4 *(in code).* Price the matching put three ways: QuantLib, parity, and the forward-measure paths.

Solution 9.4, from the notebook

```
ql_put = vas.discountBondOption(ql.Option.Put, K, T1, T2)
parity = ql_call - (P05 - K * P01)
mc_put = P01 * np.maximum(K - bond_f, 0)
se = mc_put.std() / math.sqrt(n_paths)
assert abs(ql_put - parity) < 1e-10 and abs(mc_put.mean() - ql_put) < 3 * se + 2e-5
print(f"put: QuantLib {ql_put * 1e4:.2f} bp, parity {parity * 1e4:.2f} bp, forward-measure
Monte Carlo {mc_put.mean() * 1e4:.2f} ± {se * 1e4:.2f} bp")
```

What it prints. `put: QuantLib 67.09 bp, parity 67.09 bp, forward-measure Monte Carlo 67.48 ± 0.31 bp`

Exercise 9.5 *(in code).* Use the 2-year forward measure: simulate to $t = 2$ with drift $a(\theta - r) - \sigma^2 B(t, 2)$ and check that the average of $P(2, 5)$ is $P(0, 5)/P(0, 2)$.

Solution 9.5, from the notebook

```
rng2 = np.random.default_rng(11)
x = np.full(50_000, R0); h = 2 / 504
for i in range(504):
    Bt = (1 - math.exp(-A * (2 - i * h))) / A
    x = x + (A * (THETA - x) - SIG ** 2 * Bt) * h + SIG * math.sqrt(h) *
rng2.standard_normal(x.size)
b25 = vasicek_bond(3.0, x, A, THETA, SIG)
target = P05 / float(vasicek_bond(2.0, R0, A, THETA, SIG))
se = b25.std() / math.sqrt(x.size)
assert abs(b25.mean() - target) < 3 * se + 2e-5
print(f"E^2[P(2,5)] = {b25.mean():.5f} ± {se:.5f}; P(0,5)/P(0,2) = {target:.5f}")
```

What it prints. `E^2[P(2,5)] = 0.89754 ± 0.00009; P(0,5)/P(0,2) = 0.89753`

Checkpoint, glossary and further reading

Part 3 checkpoint

Try these before Part 4. The answers follow.

1. Solve $dX = -aX\,dt + \sigma\,dW$ with $X_0 = x$. What are $E[X_t]$ and $\mathrm{Var}(X_t)$?
2. A share has drift 10% and volatility 30%, and the risk-free rate is 4%. What is the market price of risk, and the Girsanov weight Z_T?
3. Write the Feynman-Kac PDE for $u(t,x) = E[g(X_T) \mid X_t = x]$ with $dX = \kappa(m - X)\,dt + s\,dW$ and no discounting.
4. An up-and-in call and an up-and-out call have the same strike and barrier. What do their prices add up to?

Answers.

1. With the integrating factor e^{at}: $X_t = xe^{-at} + \sigma\int_0^t e^{-a(t-s)}\,dW_s$, so $E[X_t] = xe^{-at}$ and $\mathrm{Var}(X_t) = \sigma^2(1 - e^{-2at})/(2a)$ (Chapter 2).
2. $\theta = (10\% - 4\%)/30\% = 0.2$, and $Z_T = \exp(-0.2\,W_T - 0.02\,T)$ (Chapter 5).
3. $u_t + \kappa(m - x)\,u_x + \frac{1}{2}s^2 u_{xx} = 0$, with $u(T,x) = g(x)$ (Chapter 7).
4. The vanilla call: on every path exactly one of the two is alive at expiry (Chapter 8).

Glossary

Affine model. A short-rate model whose bond prices are $A(\tau)e^{-B(\tau)r}$; Vasicek and CIR are affine.

Barrier option. An option that starts (knock-in) or stops (knock-out) existing if the underlying reaches a barrier.

Brownian bridge. Brownian motion conditioned on its value at the end of an interval.

CIR model. $dr = a(\theta - r)\,dt + \sigma\sqrt{r}\,dW$ (Cox, Ingersoll and Ross).

Complete market. One in which every payoff can be replicated by trading, so it has one price.

Euler (Euler-Maruyama) scheme. $X_{n+1} = X_n + \mu\,\Delta t + \sigma\,\Delta W_n$.

Feller's condition. $2a\theta \geq \sigma^2$: a CIR process never reaches zero.

Feynman-Kac theorem. A discounted expected payoff solves $u_t + \mu u_x + \frac{1}{2}\sigma^2 u_{xx} - ru = 0$.

Forward measure. The measure with a zero-coupon bond $P(t,T)$ as numéraire; prices in units of that bond are martingales.

Geometric Brownian motion. $dS = \mu S\,dt + \sigma S\,dW$; $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$.

Girsanov's theorem. Under $dQ/dP = e^{-\theta W_T - \frac{1}{2}\theta^2 T}$, $W_t + \theta t$ is a Brownian motion.

Half-life. The time for the expected gap to the long-run level to halve: $\ln 2 / a$.

Lipschitz condition. $|f(x) - f(y)| \leq K|x - y|$; with linear growth, it guarantees a unique strong solution.

Market price of risk. $(\mu - r)/\sigma$: extra drift per unit of volatility.

Martingale representation theorem. Every square-integrable Brownian martingale is $M_0 + \int H\,dW$.

Milstein scheme. Euler plus $\frac{1}{2}\sigma\sigma'((\Delta W)^2 - \Delta t)$.

Novikov's condition. $E[e^{\frac{1}{2}\int_0^T \theta^2 dt}] < \infty$: enough for Girsanov with a random θ.

Numéraire. A positive traded asset used as the unit of value.

Ornstein-Uhlenbeck process. $dX = a(\theta - X)\,dt + \sigma\,dW$.

Reflection principle. $P(\max_{t \leq T} W_t \geq m) = 2P(W_T \geq m)$.

Running maximum. $M_T = \max_{t \leq T} W_t$.

SDE (stochastic differential equation). $dX = \mu(t, X)\,dt + \sigma(t, X)\,dW$, meaning the corresponding Itô integral equation.

Strong and weak error. $E|X_N - X_T|$ (paths) and $|E[g(X_N)] - E[g(X_T)]|$ (averages).

Vasicek model. The Ornstein-Uhlenbeck process as a short-rate model.

Further reading

This book is written from its own notebooks; these are excellent companions. None of their examples or exercises are used here.

- Steven E. Shreve, *Stochastic Calculus for Finance II: Continuous-Time Models*: Chapter 5 (Girsanov, martingale representation) for Chapters 5 and 6, Chapter 6 (Feynman-Kac) for Chapters 1 and 7, Chapter 7 (barriers) for Chapter 8, Chapters 9 and 10 (numéraires, term structure) for Chapters 2, 3 and 9.
- Bernt Øksendal, *Stochastic Differential Equations*: Chapter 5 (existence and uniqueness) and Chapter 8 (Girsanov, Feynman-Kac).
- Paul Glasserman, *Monte Carlo Methods in Financial Engineering*: Chapter 6 for discretisation (Chapter 4) and barrier corrections.
- Damiano Brigo and Fabio Mercurio, *Interest Rate Models: Theory and Practice*: Chapter 3 for one-factor short-rate models.

Check the chapter pointers against the edition you have.